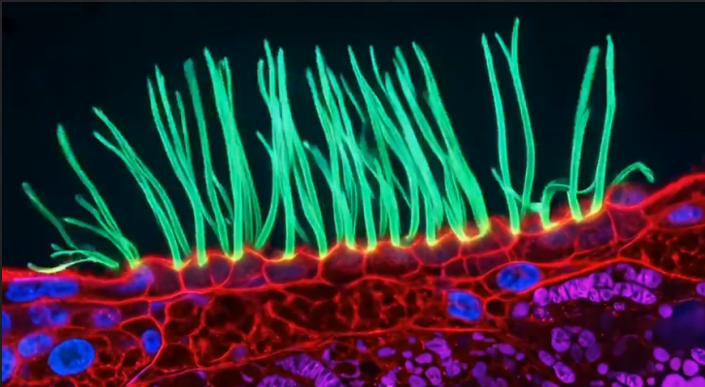


Soft-Matter @ NORDITA

Dhrubaditya (Dhruba) Mitra

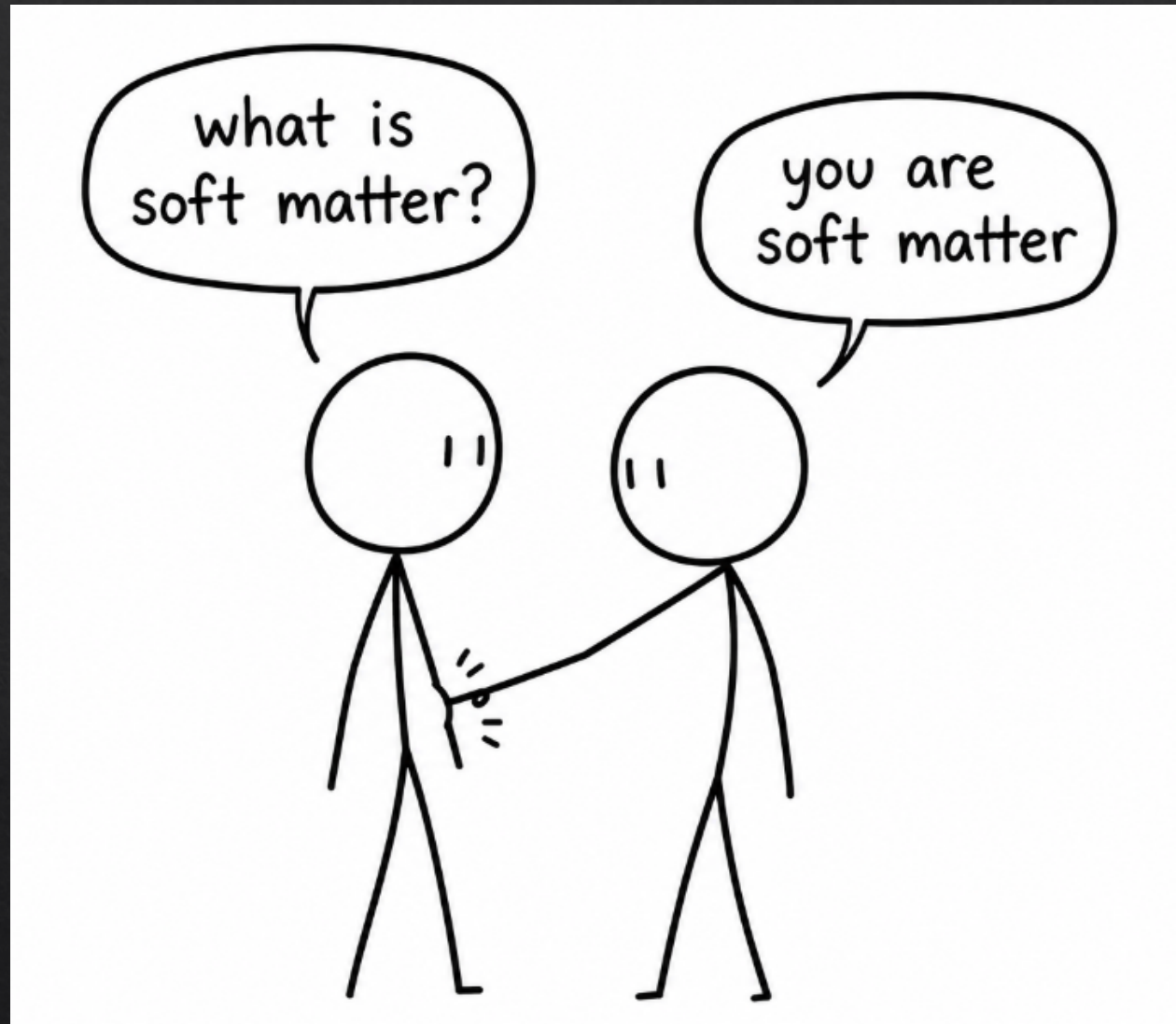


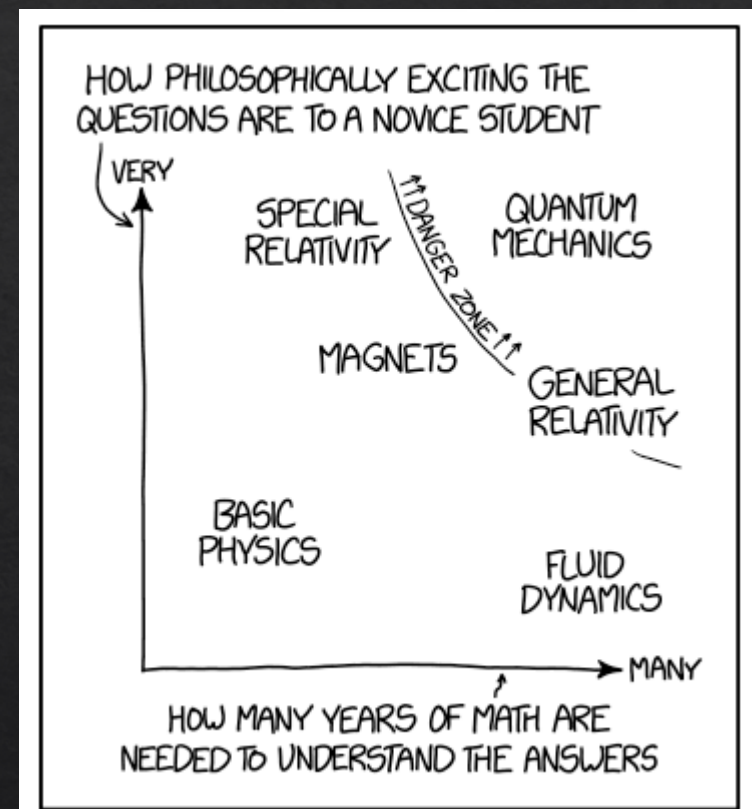
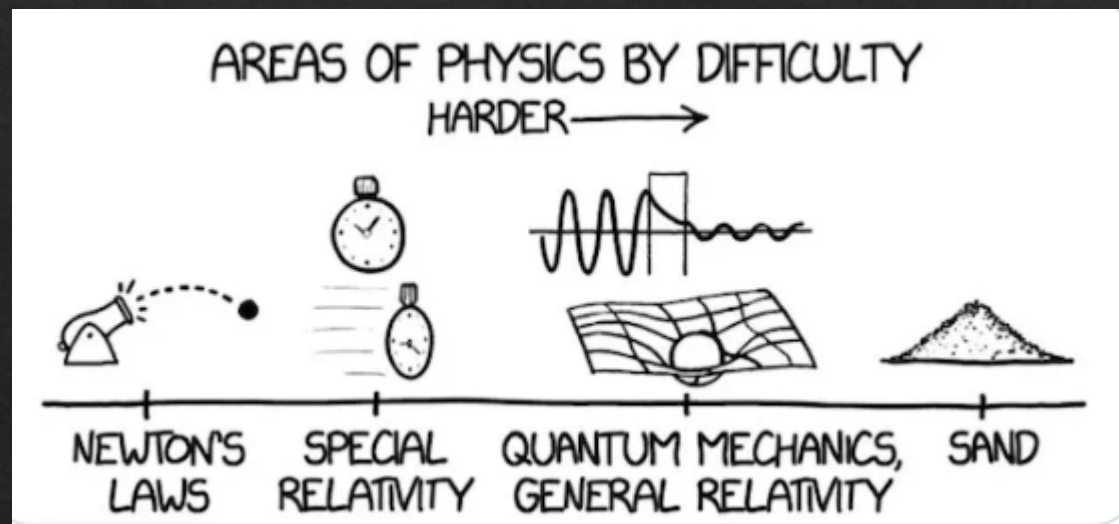
us

- Anthony Bonfils
- Ralf Eichorn
- Dhrubaditya (dhruba) Mitra
- Pijush Patra
- John Wettlaufer

friends

- Mason Porter (NORDITA)
- Supriya Krishnamurthy (SU)





Abnormal beating of airway cilia (micro meter)

<https://www.nikonsmallworld.com/galleries/2026-small-world-in-motion-competition/abnormal-beating-of-airway-cilia-from-a-child-with-primary-ciliary-dyskinesia-pcd>

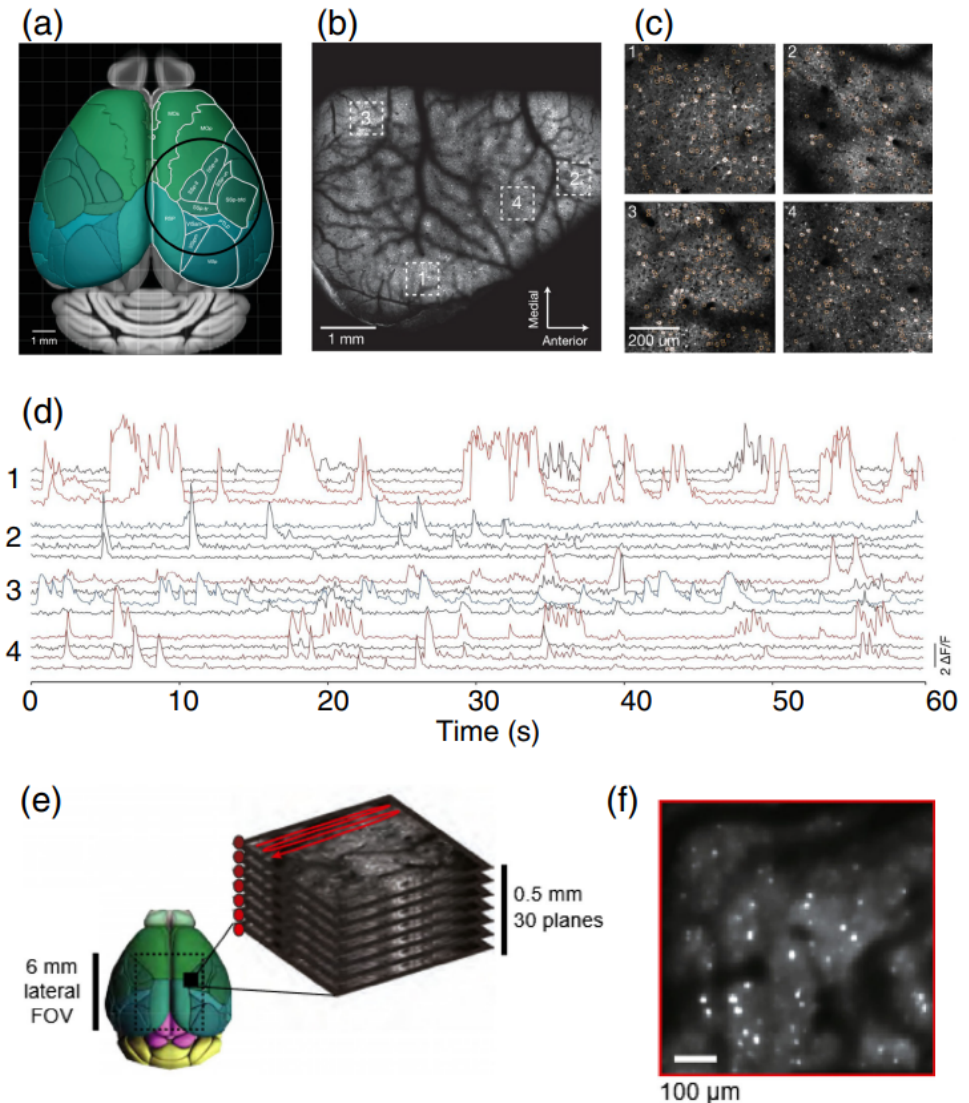


3D Medical Animations : John Leiber : Art of the Cell

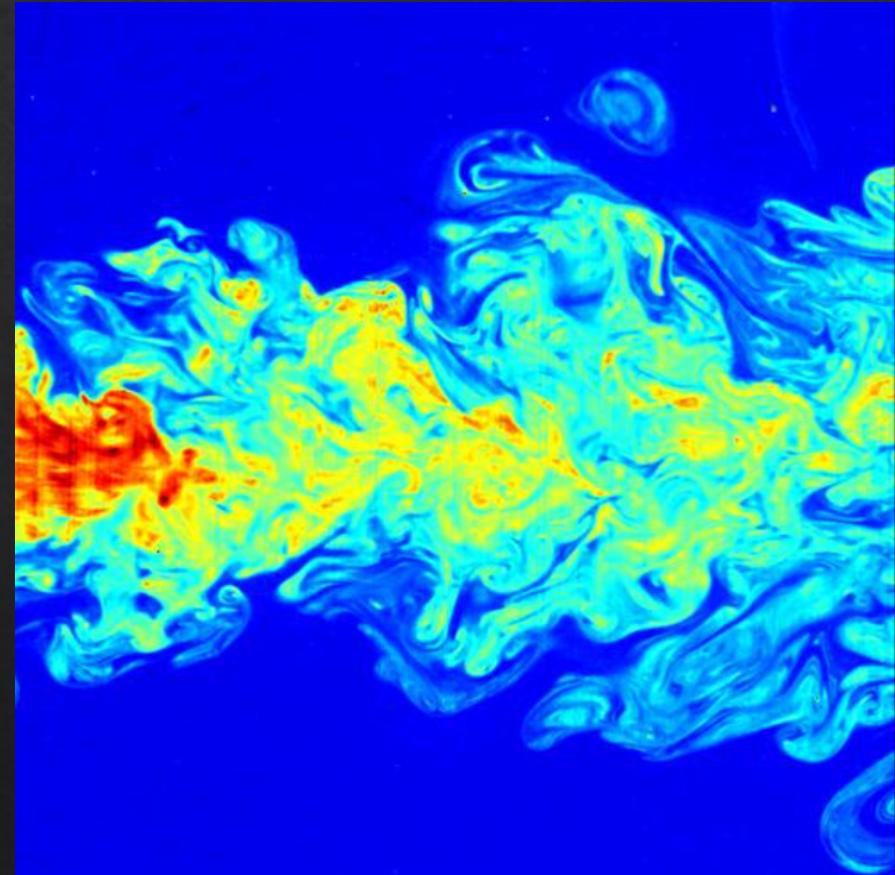
Animation of subcellular processes

Imaging from individual cells with voltage sensitive dyes (scale mm)

Data from mouse cortex. Sampling rate is 9.6 Hz. Individual neurons can sometime be identified and If we know the underlying wiring diagram then the activity can be connected to that. In two dimensions we are reaching 1000 neurons, the recording can go on for days. There are challenges to make sure that the recordings are from the same cell.



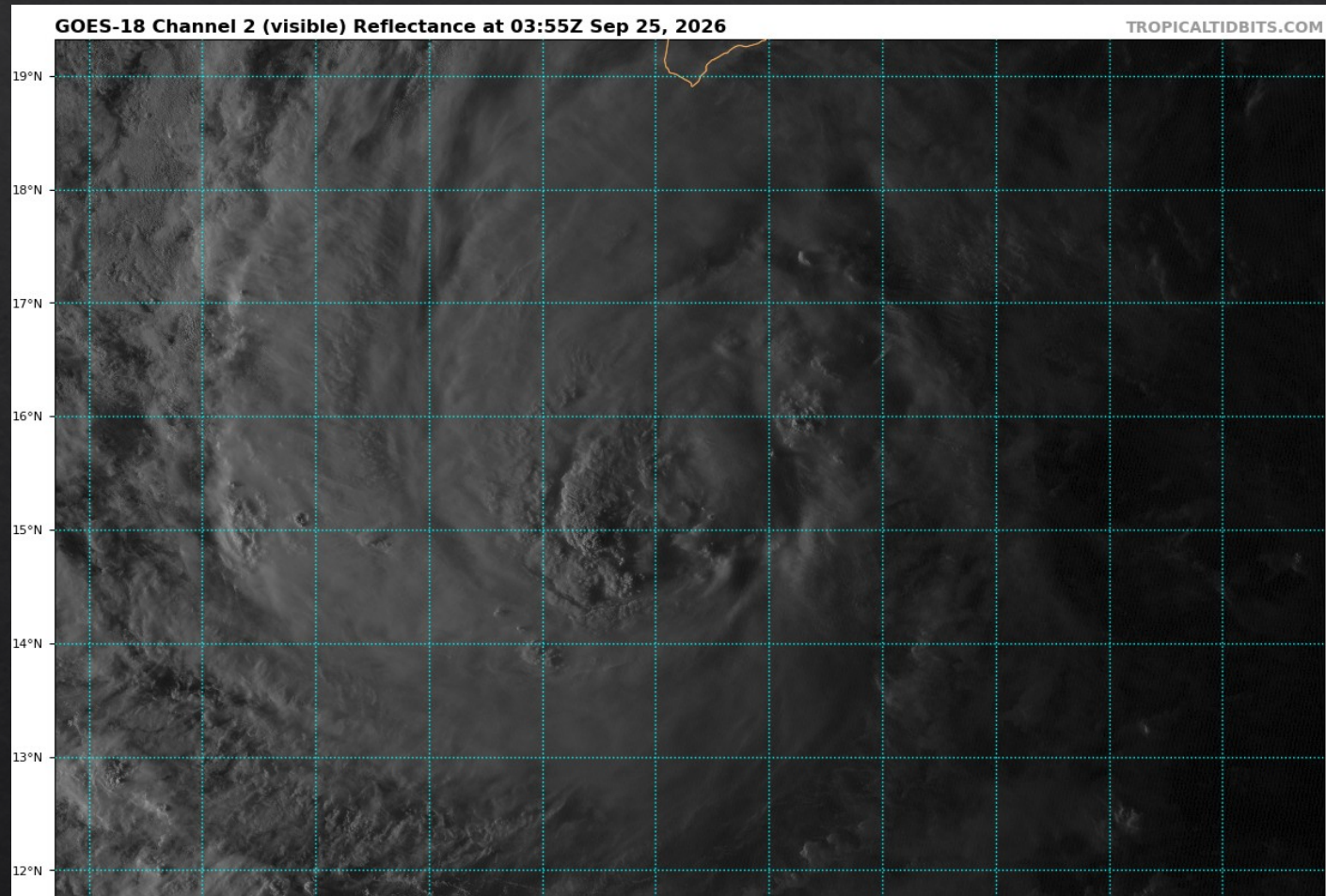
Engineering Turbulence (scale meters)



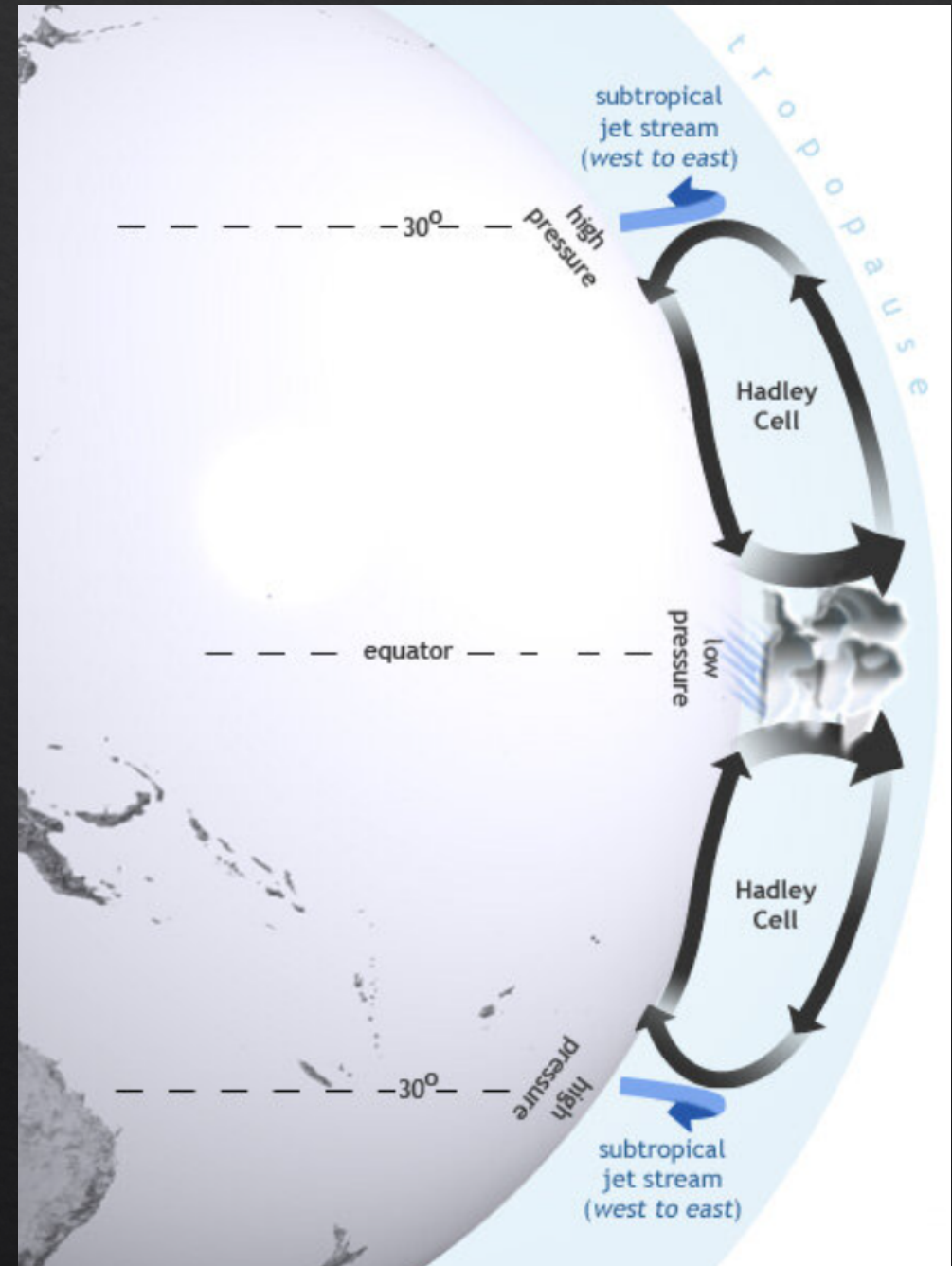
Ice floe (scale 10 meter)



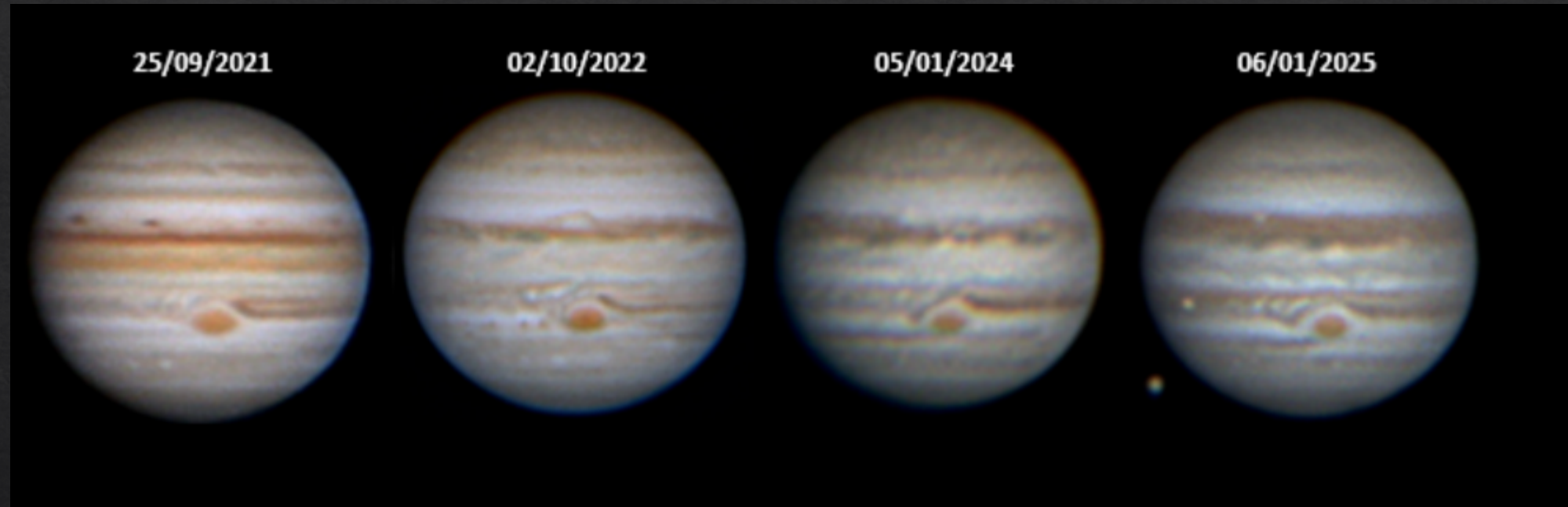
Tropical cyclone (scale kms)



Hadley cells (scale 100 km)



Great red spot of Jupiter (scale Mm)

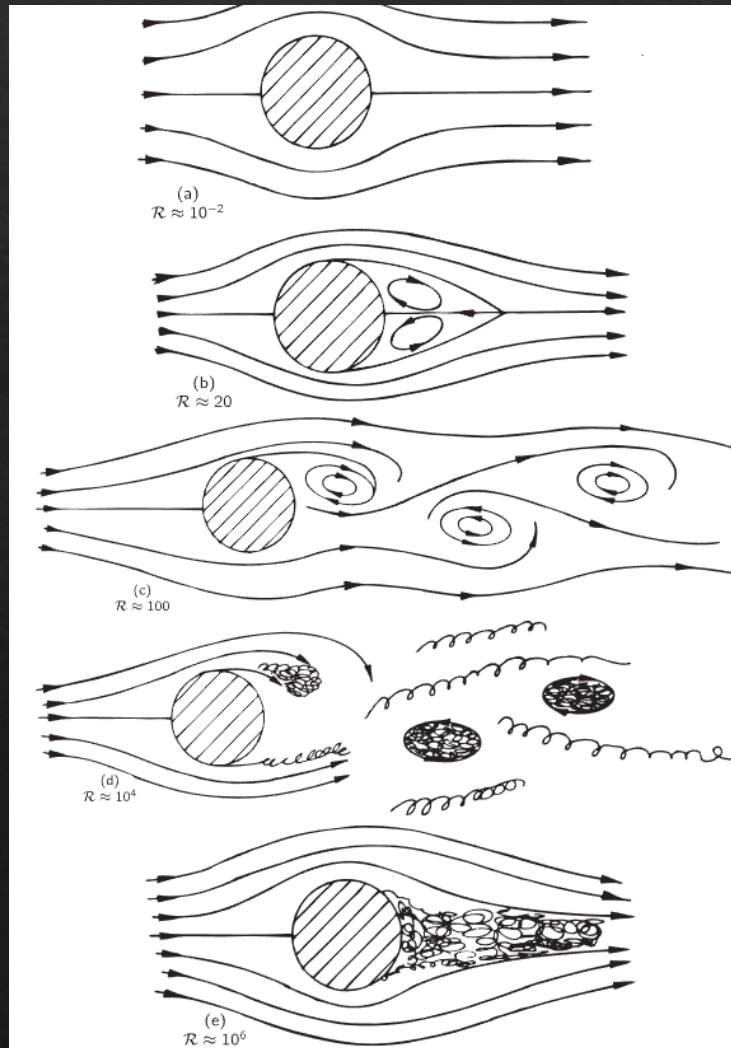


What do we do ?

- Statistical Mechanics (Beyond second law, active matter ...)
- Biophysics
- Climate Science
- Turbulence
- Fluid Mechanics
- Elasticity
- Neuroscience
- Astrophysics

Why is it difficult ?

“ I was born late, all easy problems are solved, all beautiful women are married ” - Landau



NON-LINEAR

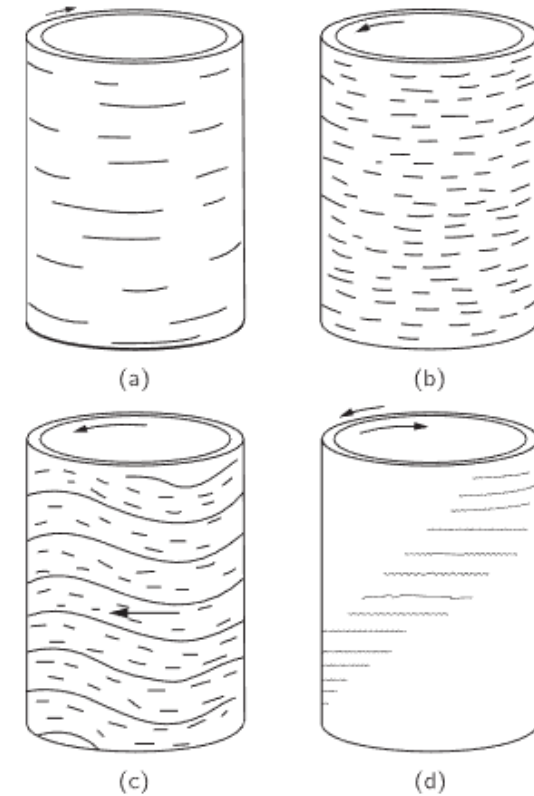
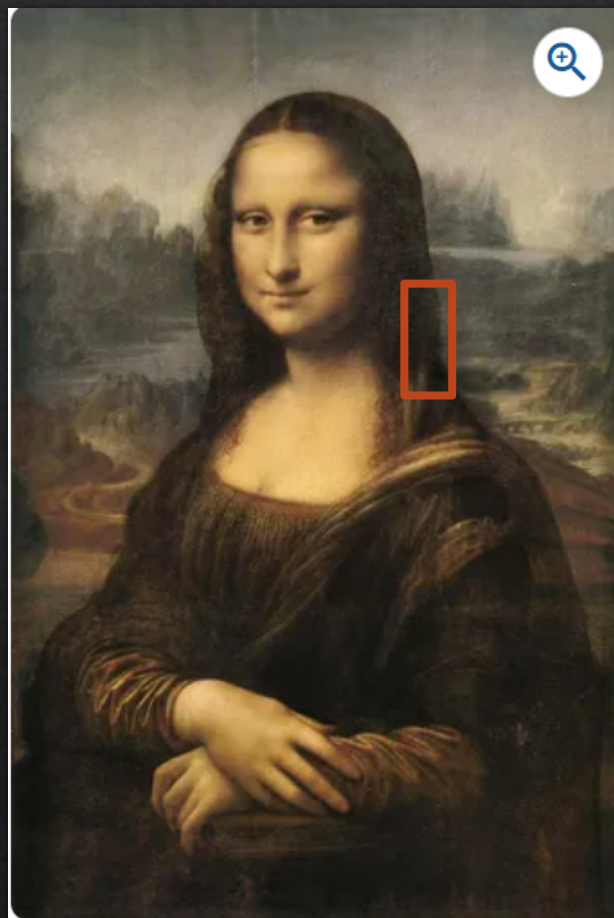


Fig. 41-8. Liquid flow patterns between two transparent rotating cylinders.

Intermediate asymptotics



There is a range of scales over which the theory is scale invariant

NON-EQUILIBRIUM

Thermodynamics

2nd law

$$\langle \Delta S \rangle \geq 0$$

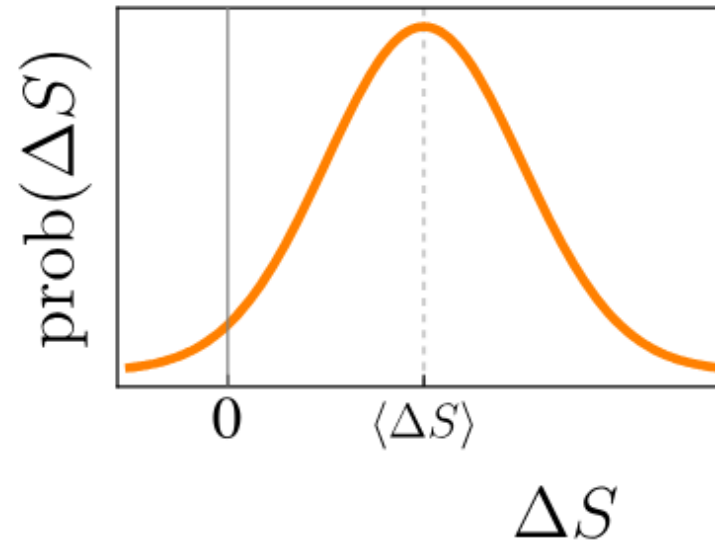


- forbids **perpetual motion machines**
- limits the efficiency of heat engines
- thermodynamic arrow of time
- ...

Stochastic Thermodynamics

refinement of the 2nd law

$$\langle e^{-\Delta S} \rangle = 1$$



e.g. for Brownian motion

↪ **thermodynamic consequences?**

Techniques

- Numerical Simulations (Monte-carlo, PDEs, ML/AI, GPU computing ...)
- Probability Theory
- PDEs
- Asymptotic Methods (matched expansions, boundary layer, renormalization group ...)
- Stability theory

Chaos is brickyard

- Q: How to make sense out of so much data ?
- Counter Q : how is it done in other parts of physics ?

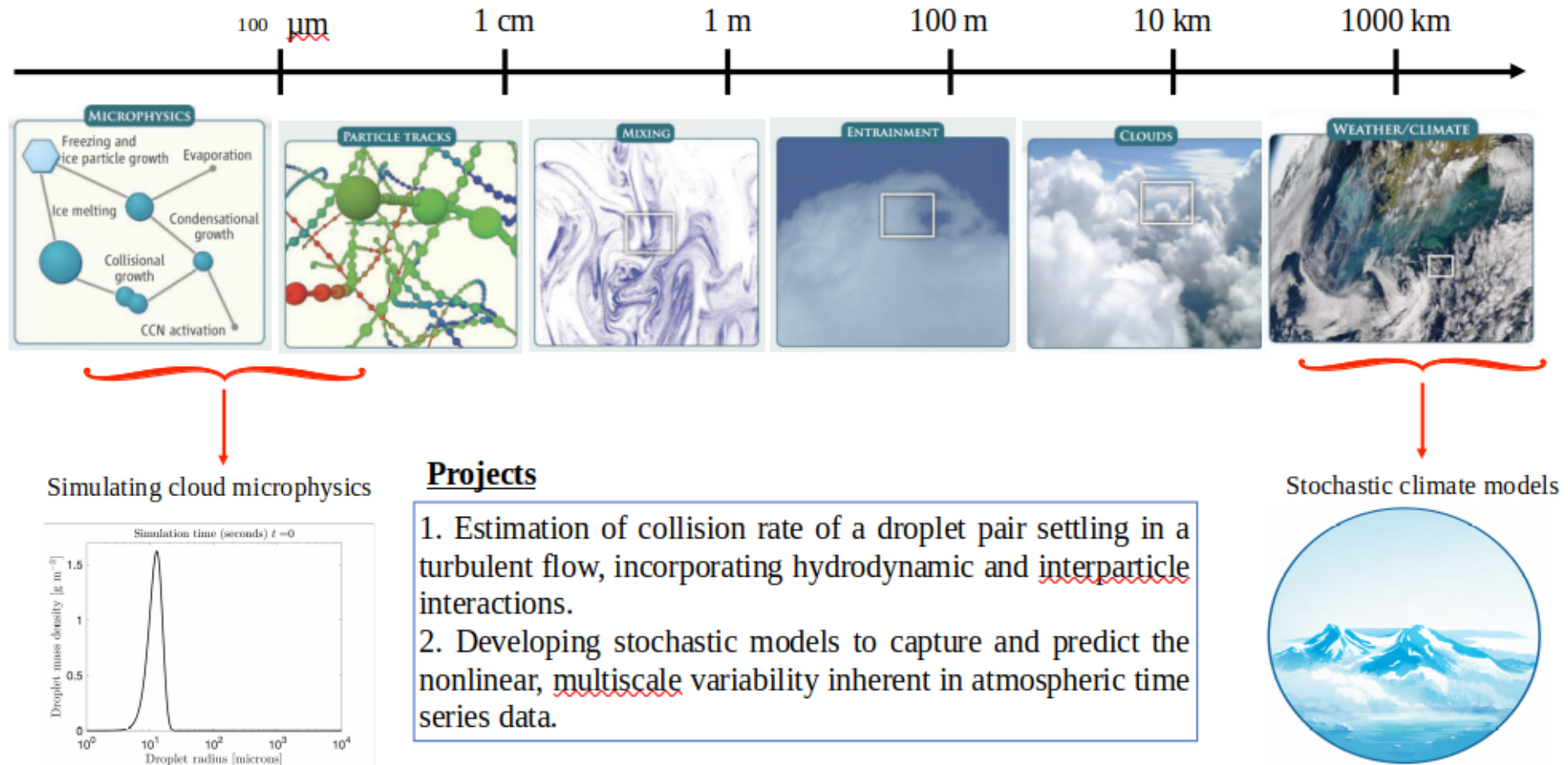


What do want from you

- Unlimited curiosity
- Excellent work ethics
- Supriya's course on stat mech. (SU students)
- Stat. Mech., Fluids courses for KTH students
- Computational Physics students welcome
- Want to apply QFT/RG to soft matter problems ? Courses on path integrals and Feynman Diagrams either in QFT or Stat Mech

Pijush Patra

Multiscale, multiphysics, and data-driven Modeling of geophysical systems



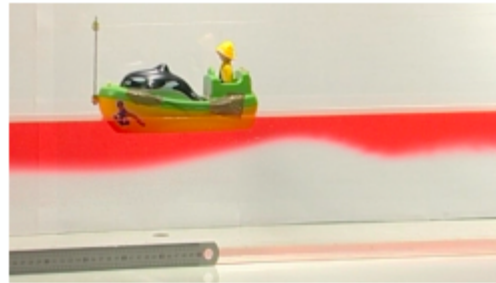
Wind Waves



Wakes and Kelvin Waves



Resurrecting Dead Water



Gunwale Bobbing



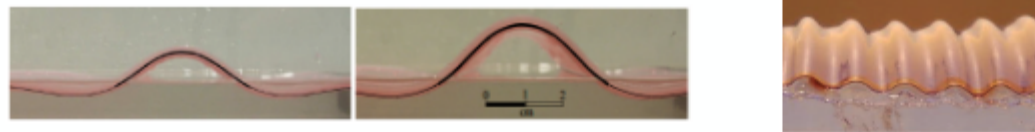
Anthony Bonfils Applied Mathematics



How should you cook a steak ?



How frequently should you flip it ?



Wrinkling and Blistering of Thin Elastic Sheets

